

LAWBOT: A Smart User Indian Legal Chatbot using Machine Learning Framework

Nikita
Dept. of CSE, SVNIT
Surat, India
nikitaniki027@gmail.com

Esha Srivastav
Dept. of CSE, SVNIT
Surat, India
esha1238@gmail.com

Aasthaben Patel
Dept. of CSE, SVNIT
Surat, India
aasthapatel1811@gmail.com

Anjali Singh
Dept. of CSE, SVNIT
Surat, India
anjusingh2810@gmail.com

Riya Sharma
Dept. of CSE, SVNIT
Surat, India
ilusharma0602@gmail.com

Dipti P Rana
Dept. of CSE, SVNIT
Surat, India
dpr@coed.svnit.ac.in

Rupa G Mehta
Dept. of CSE, SVNIT
Surat, India
rgm@coed.svnit.ac.in

Abstract—Legal documents and articles in India are abundant on the Internet and in libraries, making it a tedious task for the average person unfamiliar with the legal field to wade through. They need proper guidance, be it filling out forms, understanding legal terms of service, filing a complaint, or any other legal activity. Developing a simple and efficient chatbot that utilizes the complex dataset and uses natural language processing to provide information about applicable laws and documents is essential. We propose Machine Learning framework based LAWBOT which aims to create a conversational chatbot for these users to easily ask legal questions, find lawyers, and review lawsuit documents.

Index Terms—Indian Legal Chatbot, Legal Stopwords, Machine Learning framework, RASA, Embeddings

I. INTRODUCTION

Legal documents in India are abundant online and in libraries, making navigation difficult for the average citizen. Legal chatbots can provide easy access to legal information and advice. The conversational chatbot can help users ask legal questions, find lawyers, and review lawsuit documents. Chatbots are accessible to all societal categories, require no special training, and are efficient, scalable, and adaptable, making them ideal for organizations providing legal services to a large number of people.

Legal chatbots have the potential to revolutionize the way that legal services are delivered in India. They can help to make the law more accessible and affordable for ordinary citizens. As the legal chatbot market in India continues to grow, it is important to ensure that these technologies are used responsibly and ethically and people have access to accurate and unbiased legal information. With careful planning and implementation, legal chatbots can play a significant role in improving access to legal information in India.

Legal chatbots can also help to improve the efficiency of the legal system by automating routine tasks, increase transparency in the legal system by making it easier for people to access information about the law and promote civic engagement by encouraging people to learn more about their legal rights and responsibilities. Overall, legal chatbots have

the potential to make a significant positive impact on the Indian legal system.

We use Machine Learning based RASA framework to build a conversational chatbot called LAWBOT. This work aims at developing a simple and efficient chatbot that utilizes the complex dataset and uses Natural Language Processing (NLP) to provide information based on requirements. The chatbot provides functionalities like fetching case documents based on choices and keywords provided by user, lawyer information according to the requirements and exploring frequently asked legal questions. Fetching documents based on choices makes use of embeddings and cosine similarity. Lawyer information search applies the concept of pattern matching and searching frequently asked questions takes advantage of inbuilt RASA function tool for pattern mapping.

The rest of the paper is organized in 4 sections. Section 2 includes work done involving legal chatbot, Section 3 shows the details of proposed work. Section 4 shows detailed implementation analysis and Section 5 includes conclusion and future work.

II. LITERATURE SURVEY

It has always been difficult to find legal information and respond to legal inquiries. As the availability of legal documents on the World Wide Web and other specialised systems explodes, this subject is gaining more and more attention. Because of the significant under-utilization of resources and the possibility of information quality problems, this expansion is not being matched by a corresponding increase in information analysis skills. The literature review of legal chatbots and legal information retrieval is therefore covered in this section.

Vorada Socratyanurak et al. [1] developed a chatbot for Thai sexual violence victims and survivors using 182 Supreme Court case data, SpaCy for similarity scores, TF-IDF algorithm for common keywords extraction, and Wordnet corpus for synonymous words search. Soha Khazaeli et al. [2] developed a legal question answer system using sparse vector search and embedding for input. The system uses BM25, Siamese

Legal BERT, and Legal GloVe for querying and searching. The "answer finder" reranks the top 30 passages and selects the top three, evaluated by Subject Matter Experts on a scale of -1 to 3. This method improves performance by filtering out unrelated passages.

Shubhashri G et. al. [3] developed an AI-driven bot technology to provide case information based on various fields, using the Dynamic Memory Network (DMN) Architecture. The model generates an abstract summary of judgment papers using a neural network-based sequence to sequence model, with the top five documents chosen using Word Mover’s Distance (WMD). Firdaus et al. [4] utilized an intelligent chatbot to analyze user queries for Indonesian law on electronic information and transactions using natural language processing. The algorithm translated non-standard keywords into standard ones, analyzed user input, and used various tools for accurate results. Lex2KG is a system developed by Abdurrahman et al. [5] for automatically converting legal documents into Knowledge Graphs (KG). It extracts relationships among legal resources, metadata, structure, and textual content, and presents the data as a legal KG.

To solve RTE in civil law, Kim et al. presented a hybrid approach that combines straightforward rules and unsupervised learning with deep language characteristics [6]. Additionally, the author created a database of negation and antonym phrases that would be used to categorise basic inquiries. The author identified premises and conclusions by using morphological, syntactic, and lexical analysis to address challenging topics. Carvalho et al. [7] proposed three steps for legal information retrieval: morphological analysis, machine learning algorithm assessment, and text replying. They used a mixed-size n-gram model to rank relevant articles, assess questions and articles, and provide correct answers. However, limitations include lack of implicit semantic representation.

Legal document retrieval and question-answering for legal inquiries were the two areas of the information processing that Kim et al. concentrated on in order to respond to yes/no questions on legal bar tests [8]. They employed TF-IDF and Ranking SVM models for the first phase. The Stanford NLP technology tokenized and processed both queries and articles. For the Information Retrieval (IR) job, the terms in a query and an article were used to determine how similar they are. For the second phase, they used a Convolutional Neural Network (CNN) model with pre-trained semantic word embeddings and dropout regularisation Rectified Linear Units and collected deep language aspects based on morphological analysis and dependency trees, including lexical and syntactic information.

Do et al. [9] developed a pipeline system for Information Retrieval (IR) and Question Answering (QA) using SVM Rank, a powerful pairwise ranking algorithm. They used a two-aspect, expanded version of Kim et al. [8] model, which used lexical words, dependency pairs, and TF-IDF score. They applied the CNN model for QA, revealing that CNN models are sensitive to parameter initial values. They ran each model n times to choose the best-optimized parameters, and found that CNN performed better with additional characteristics like

LSI and TF-IDF.

From the literature survey, it can be observed that researchers have explored legal chatbot for different legal matters around the world, it is still an open field of research for a fully functional legal chatbot in context of India. We explore some of the legal services which can be provided through Indian legal chatbot.

III. PROPOSED WORK

We propose an Indian Legal Chatbot **LAWBOT** which provides required information to the user as per the choice. Fig. 1 shows the proposed architecture for the **LAWBOT**. The details of each functionalities and creation of database is discussed in further subsections.

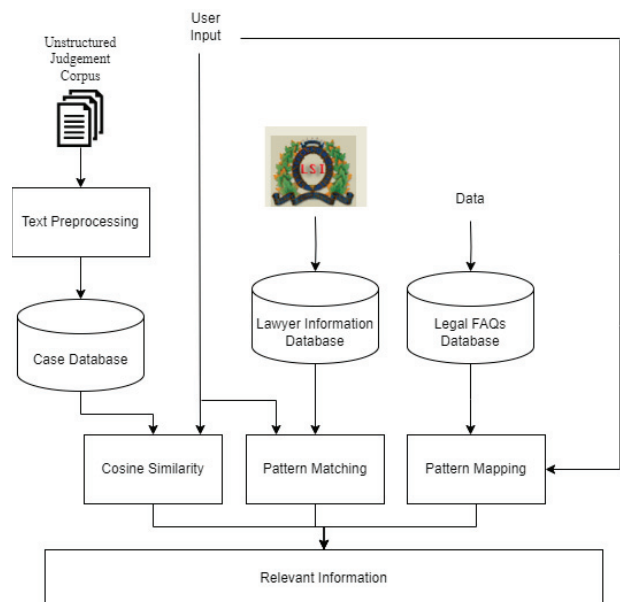


Fig. 1. Proposed Architecture for LAWBOT

Fig. 2 shows the overall workflow of the LAWBOT. It illustrates how the user will use the LAWBOT to take advantage of the functionalities and services provided.

Initially, the user will be greeted by the LAWBOT and asked to choose the type of service they require. Based on their choice, they will be assisted. The user can either choose to study a case or search for a lawyer based on their requirements or search for legal questions which are the most commonly asked questions by the civilians regarding the legal domain. Section III-A and III-B further elaborate the flow and methodology adopted for generating the given services of the LAWBOT whether user wishes to search for a case or search for a lawyer respectively. Section III-C shows the details of how legal frequently asked questions (FAQs) were created for our database using inbuilt features of RASA.

A. Case Study

This function of LAWBOT enables users to search for a particular case with the help of options provided in the menu of LAWBOT and user entered keywords. To facilitate this

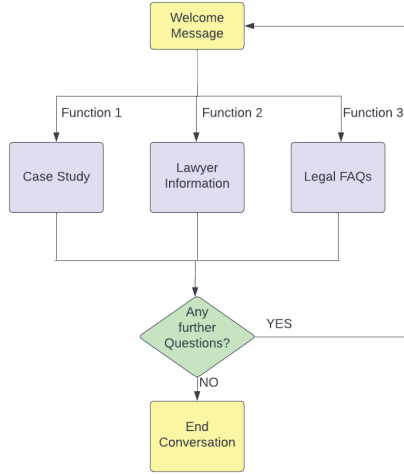


Fig. 2. Working Flow Diagram - LAWBOT

function, we have created a static database with the help of judgment documents in our dataset.

1) *Text Processing*: The documents in our dataset are pre-processed utilizing standard pre-processing techniques such as Tokenizer, WordNet and Lemmatizer and also includes removal of Legal Stopwords and generating embeddings.

Legal Stopwords: The commonly occurring legal terms like judge, lawyer, victim, accused, court, attorney, jury, etc., are present in every legal document and hence their presence can generate misleading similarity in case of non similar documents. Removing those recurring legal terms, duplicate words and numbers along with stopwords in the English language would help us get rid of such misleading similarity and shrink our data size considerably.

As depicted in Fig. 3, to generate common legal terms, we first removed English stopwords from the documents and then tokenized the remaining words from each document. Symbols $d1 \dots dn$ represents documents number $d1$ to dn from our corpus. We then generated a list of words by selecting common words from each document using intersection process. The resultant list contained the words which were common in every document, called Legal Stopwords. A sample of Legal Stopwords is shown in Table I. These legal stopwords are removed from the documents before embedding generation.



Fig. 3. Legal stopwords Generation

Embeddings: Longformer model [15] is used to generate embeddings for a given document, thereby capturing the semantic context of the text. The final output is a set of vectors, each corresponding to a token in the input sequence. This allows the model to generate meaningful word embeddings that capture contextual information across extensive text spans.

The generated embeddings are stored in the Case database to be matched with user embeddings.

2) *Cosine Similarity*: After user input is obtained, it undergoes the same tokenization process, ultimately resulting in user embeddings. The key insight here is that these embeddings serve as numerical representations of the textual content, allowing for direct quantitative comparisons. To measure the similarity between the document and user input, a cosine similarity matrix was computed using the embeddings. The cosine similarity metric ranges from -1 to 1, where a higher value indicates greater similarity. This matrix is further used to find the top three case documents closest related to the input given by the user. Table III shows the cosine similarity matrix generated for the 3 keywords given as input by the user and the word embeddings of 5 documents. The cosine similarity values shown in the table are the average of cosine similarity value between each input keyword and the words in the document. We take documents D1, D2, D3, D4 and D5 as a sample set of 5 documents. The top 3 documents are taken as most similar documents relative to the input given by the user.

B. Lawyer Information Search

The dataset was obtained from Legal Services Website using web scraping tools and a Lawyer Information Database was created. Now, based on the input of the user, keywords are matched with the information of the lawyer in the lawyer information database using pattern matching approach provided by python. The most relevant lawyer information is given to the user.

C. Legal FAQs

This functionality of LAWBOT is fulfilled using *pipeline* processing provided by RASA.

Intent Recognition: Intents are different topics on which chatbot is to be trained to generate answer for user query. We created a intent training dataset with 20 intents and corresponding set 5 examples for each intent, a sample from our intent training dataset is shared in Fig. 4. Intent represents the intention or purpose behind a message to find the most similar result relevant to user input. RASA uses the provided training data to train a machine learning model (we used sklearn library) to recognize intents. During training, Rasa processes the training data and learns patterns and features that help it understand the mapping between user inputs and intents.

Pattern Mapping: Based on user input, RASA utilizes the machine learning model to classify incoming user messages into predefined intents. The goal is to determine which intent(s) best represent the intention of the user input based on the patterns learned during training. The intent pattern mapping process assigns a similarity score to each intent based on how closely the input message aligns with the patterns learned during training. RASA considers the intent with the highest similarity score as the predicted intent for the user message.

TABLE I
LEGAL STOPWORDS

Legal Stopwords	Supreme, high, court, judge, lawyer, attorney, counsel, plaintiff, defendant, witness, evidence, hearsay, objection, jury, verdict, appeal, testimony, deposition, affidavit, allegation, complaint, counterclaim, cross-claim, answer, discovery, motion, pleading, brief, petition, citation, statute, regulation, case, decision, opinion, legal, jurisdiction, respondent, petitioner, judgment, hearing, verdict, section etc.
-----------------	---

TABLE II
CASE DATABASE

Case Name	State	Case Link	Word Embeddings
Gian Singh vs State Of Punjab & Anr on 24 September, 2012	Supreme	https://indiankanoon.org/doc/69949024/	[array([-2.64901575e-02, 1.02907814e-01, -1.27305742e-02, -3.52696925e-02, 8.30573142e-02, -8.97115469e-02, -2.78878417e-02, 4.43472862e-02, 6.60640225e-02, -9.97927189e-02, 4.18482944e-02,
Sarla Verma & Ors vs Delhi Transport Corp.& Anr on 15 April, 2009	Supreme	https://indiankanoon.org/doc/837924/	[array([-2.54536923e-02, 8.53583664e-02, -1.38967335e-02, -6.51293248e-02, 9.13570151e-02, -1.09054528e-01, -1.97001733e-02, 2.12531444e-02, 6.85455799e-02, -6.39452562e-02, 3.07814069e-02,....
Rangappa vs Sri Mohan on 7 May, 2010	Supreme	https://indiankanoon.org/doc/150051/	[array([-6.71354309e-02, 8.71304795e-02, 5.66098047e-03, -7.39205852e-02, 1.16565898e-01, -8.18985701e-02, -2.20370851e-02,

TABLE III
COSINE SIMILARITY MATRIX FOR RETRIEVING RELATED CASE DOCUMENTS

	D1	D2	D3	D4	D5
hotel	0.5	0.2	0.006	0.1	0.3
demolish	0.8	0.7	0.3	0.09	0.6
ministry	0.4	0.9	0.05	0.04	0.5

```

- intent: hindu_marriage
examples: |
- women childhood marriage
- what if women is forced to marry before turning 18
- what if marriage before 18 years age for women
- what are rights for underage women who is forced into marriage
- what can i do if i am married against my will

- intent: consumer_rights
examples: |
- consumer rights
- what are consumer rights
- what are my rights as a consumer
- rights entitled to me as a consumer
- what can i do as a consumer

- intent: copyright
examples: |
- what are copyrights for a owner
- copyrights in india
- moral rights for copyright owner
- what rights do i get as a copyright owner
- what is copyright

```

Fig. 4. Intents defined for Categorizing User Queries

IV. IMPLEMENTATION ANALYSIS

A. Data Set

1) *Case Database*: After performing all these preprocessing, the collection of resultant documents forms the Judgment Database used for fetching case studies details. Table II shows a sample from the Judgment database created after applying preprocessing techniques and generating embeddings. This database is used for information retrieval according the requirements of the user.

2) *Lawyer Information Database*: The data on the website [14] is present city wise for both High Court and Supreme

Court. The extracted dataset, includes the following fields taken from the site for each Lawyer or Law Firms in multiple states – Name of the Lawyer/Lawyer Firm, State, Address, Court of Practice and Area of Practice – using webscraping and the output is stored in the lawyer database. Table IV shows a sample lawyer data extracted from the website. We omit address field in the lawyer information database table and in the implementation analysis in later sections due to privacy.

3) *Legal FAQs Database*: Legal FAQs database contains the set of questions which were used for intent training, after intent classification user query is mapped with one of the examples in the selected intent.

B. Technologies used

The technologies that are used for the implementation till now are:

- RASA is a machine learning framework, which is a tool that uses Python and natural language understanding(NLU) to create customized AI chatbots and enables user to train the model and create unique actions. Rasa-built chatbots have been deployed on a variety of platforms, including Facebook Messenger, Microsoft Bot, Slack, and others.
- A prototype of the chatbot had been developed using Botpress which gave an overview on the flow of the chatbot and the functionality and services it performs. The chatbot is implemented to include the overall functionality of having a trained model that will allow the user to type their queries. These queries will be read by the trained model to give the most accurate answers.
- For the backend implementation, web scraping libraries in python like Requests and BeautifulSoup are used to generate the lawyer information. For preprocessing of judgement samples, Python libraries like NLTK and numpy are used.

TABLE IV
LAWYER INFORMATION DATABASE

Name	State	Court of Practice	Area of Practice
Advocate1	Maharashtra	supreme court, district court	bankruptcy-creditors, bankruptcy-debtors, collection/creditors, construction, contract, corporation / partnership, finance, intellectual property, lemon law, probate, real estate, securities, arbitration, wills, trusts, estates
Advocate2	West Bengal	calcutta high court, city civil court.	criminal, divorce/domestic relations, family law, landlord/tenant, probate, real estate, wills, trusts, estates
Advocate3	Punjab	Punjab high court, district court firm	appellate practice, arbitration, mediation, child custody, civil litigation, construction, consumer law, contract, copyright, corporation / partnership, criminal, divorce/domestic relations, entertainment, family law, intellectual property, landlord/tenant, libel/slander

- For creating word embeddings for case study keywords, Longformer model is used and cosine similarity for calculating similarity scores.

C. LAWBOT Working

The chatbot has been implemented on Rasa Framework to include all the required functionalities. On startup, the user is greeted by the chatbot and asked to select the service they require. There are 3 functionalities and they are as follows -

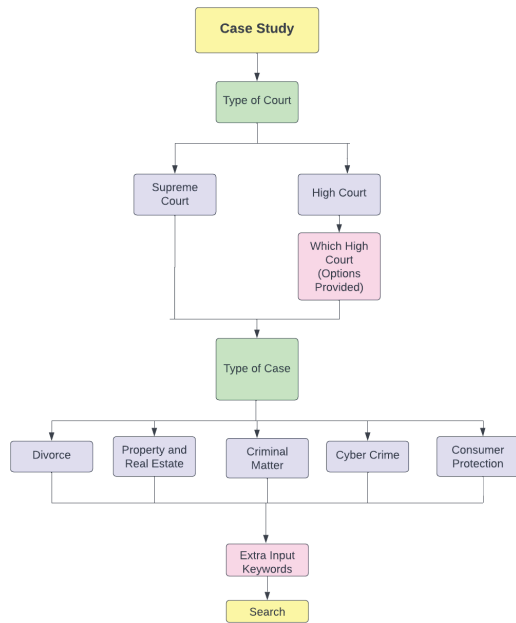


Fig. 5. Flow Diagram - Case Study

- 1) Case Study - Fig. 5 shows the workflow of the LAWBOT for the case when the user wishes to search for a case whose verdict has been given by the court. The user is prompted to choose the type of court they wish to search in where they can choose the Supreme Court or the High Court. If the user selects High Court, they are further prompted to select High Court based on the State. The user also has the option to skip selecting a court if they are not sure. Next, the user has to select the type of case they wish to search. These cases may be belonging to different domains such as civil cases, criminal cases, cyber cases, consumer cases, etc. Lastly,

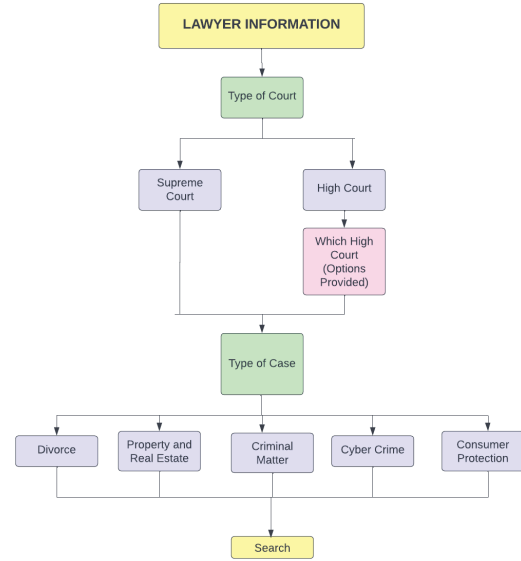


Fig. 6. Flow Diagram - Lawyer Information

the user will add the keywords regarding the case they wish to search. The LAWBOT will provide the user with the most similar cases based on their query. Based on these inputs, the data will be extracted from the Judgment database. Fig. 9 shows the results as the links to different case files which the user will get.

- 2) Lawyer Information - Fig. 6 illustrates the flow of the user when they wish to search for a lawyer. First, the user will choose the type of court they wish to pursue. The user has the option to select from either the High Court or the Supreme Court. If the user selects the High Court, they are further prompted to choose the state of the High Court. Next, the user will be prompted to select the type of case based on their requirement. The chatbot will return the lawyer whose practice is most similar to users query. Fig. 7 shows the options provided to the user in order to search for lawyer information, this information will be used to extract data from lawyer database to find the lawyer information closest to the users requirements. As depicted in the Fig. 8, the user will be given the lawyer name, contact details and area of practice as output.
- 3) Legal FAQs - In case the user wants to ask a legal

question related to the laws and the legal system, they would choose the ‘Legal FAQs’ option. The user can type their question to search. There is a set of general legal questions present in the database. The chatbot will analyze the question and try to answer as accurately as possible using RASA from the available database. This database can be scaled further to include more information. Example of such legal questions is shown in the Fig. 10.

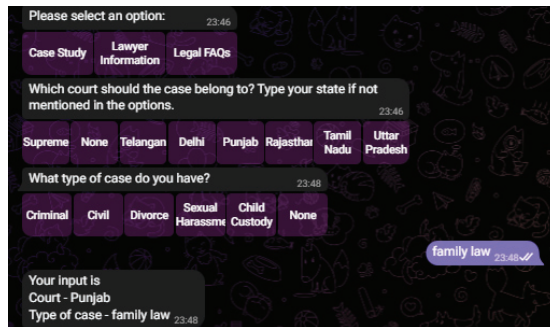


Fig. 7. Asking for lawyer information from user

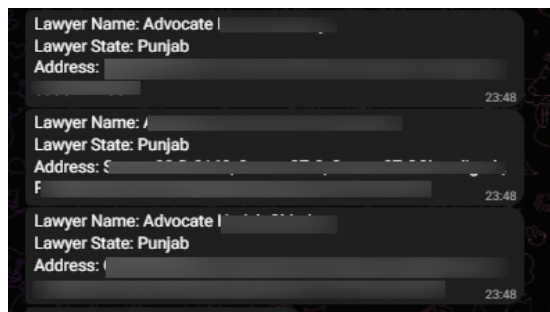


Fig. 8. Lawyer information as output

V. CONCLUSION

This work is an attempt to ease the information retrieval from the judicial system to provide a user friendly, easy to access, cost and time effective platform for the common people. Processed judgment documents are used as a database for retrieving case information in case study functionality of the chat bot. Lawyer information is extracted using web scrapping and a lawyer database is created which serves as the repository for the chat bot to provide lawyer information. Legal questions on a set of topics are presented to the user for information. For improved user interaction and experience, the chat bot is integrated with Telegram.

Since, a fully functional integrated chat bot has been implemented, user query search can be optimized for better results and clearly understanding the user intent. Further advancements such as understanding and answering queries in multiple languages can also be done. One can even partner with some firms and propose the idea of a video call meeting with a lawyer before actually meeting in person. This will reduce cost and commute time for both parties.

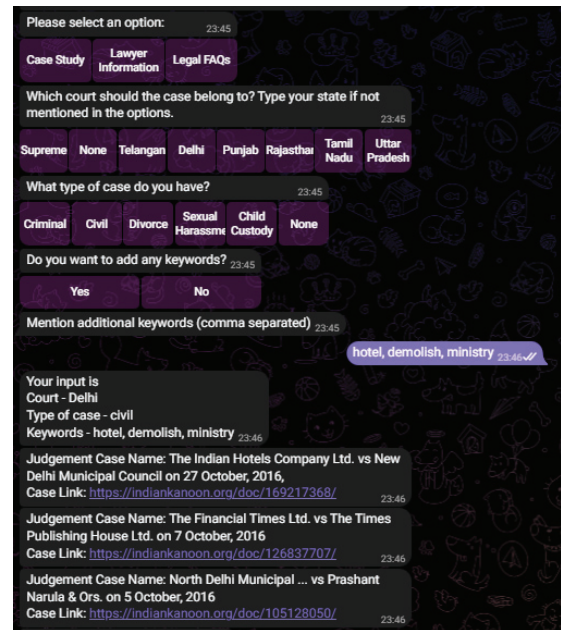


Fig. 9. Case Documents information input and output

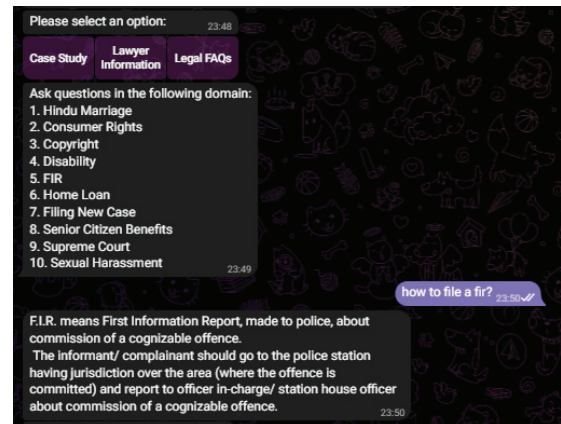


Fig. 10. Legal FAQs input and output

ACKNOWLEDGMENT

The authors would like to thank the contributors of Legal Services of India and India Kanoon for making Lawyer Information and Indian judgment documents publicly available respectively.

REFERENCES

- [1] V. Sotiyanurak et al., "LAW-U: Legal Guidance Through Artificial Intelligence Chatbot for Sexual Violence Victims and Survivors," in IEEE Access, vol. 9, pp. 131440-131461, 2021, doi: 10.1109/ACCESS.2021.3113172.
- [2] Khazaeli et al.: A Free Format Legal Question Answering System, 2021 <https://aclanthology.org/2021.nllp-1.11>
- [3] Shubhashri G, Unnamalai N, and Kamalika G. 2018. LAWBO: A smart lawyer chatbot. In Proceedings of the ACM India Joint International Conference on Data Science and Management of Data (CoDS-COMAD '18). Association for Computing Machinery, New York, NY, USA, 348–351. <https://doi.org/10.1145/3152494.3167988>
- [4] Firdaus, Vipkas & Saputra, Pramana Yoga and Suprianto, Dodit. (2020). Intelligence chatbot for Indonesian law on electronic information and transaction. IOP Conference Series: Materials Science and Engineering. 830. 022089. 10.1088/1757-899X/830/2/022089.

- [5] M. Abdurahman, F. Darari, H. Lesmana, M. Hartopo, I. Rhesa and B. C. Lumban Tobing, "Lex2KG: Automatic Conversion of Legal Documents to Knowledge Graph," 2021 International Conference on Advanced Computer Science and Information Systems (ICACSIS), 2021, pp. 1-5, doi: 10.1109/ICACSIS53237.2021.9631310.
- [6] Mi-Young Kim, Ying Xu, Randy Goebel and Ken Satoh: Answering Yes/No Questions in Legal Bar Exams". New Frontiers in Artificial Intelligence. Springer International Publishing: 199-213, 2014.
- [7] Danilo S. Carvalho, Minh-Tien Nguyen, Chien-Xuan Tran, Minh-Le Nguyen. Lexical-Morphological Modeling for Legal Text Analysis". Ninth International Workshop on Juris-informatics (JURISIN), 2015
- [8] Mi-Young Kim, Ying Xu, Randy Goebel: A Convolutional Neural Network in Legal Question Answering". Ninth International Workshop on Juris-informatics (JU-RISIN), 2015
- [9] Phong-Khac Do, Huy-Tien Nguyen, Chien-Xuan Tran, Minh-Tien Nguyen, and Minh-Le Nguyen: Legal Question Answering using Ranking SVM and Deep Convolutional Neural Network". Japan Advanced Institute of Science and Technology (JAIST), Hung Yen University of Education and Technology (UTEHY), Hung Yen, Vietnam. 2017
- [10] Open Source Chatbot Platforms: <https://botpress.com/blog/open-source-chatbots>
- [11] Natural Language Toolkit Documentation: <https://www.nltk.org/>
- [12] Tensorflow - word2vec Official Tutorial: <https://www.tensorflow.org/tutorials/text/word2vec>
- [13] Jacob Devlin, Ming-Wei Chang, Kenton Lee, Kristina Toutanova: BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding". May 2019
- [14] Legal Services of India: https://www.legalserviceindia.com/lawyers/lawyers_home.htm
- [15] Longformer: https://huggingface.co/docs/transformers/model_doc/longformer